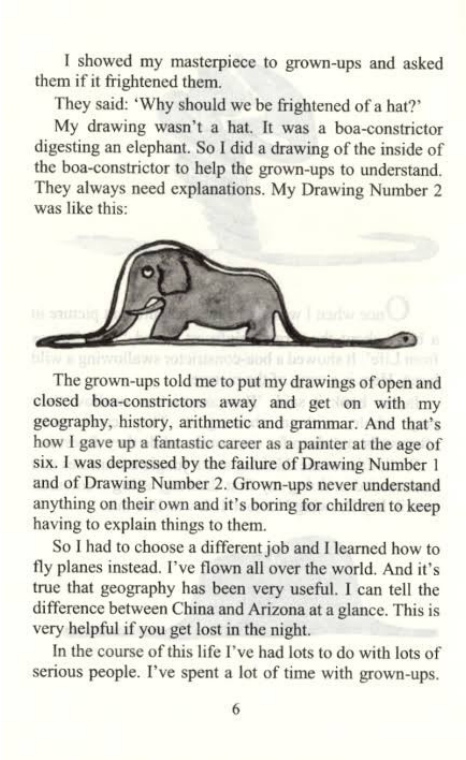


Representing Visual Concepts in Multisensory Storybooks for Blind and Visually Impaired Children

: Methods, Techniques, and Media

The visual dominance of early education creates a cognitive gap for blind and visually impaired children

Sighted Children's Materials



I showed my masterpiece to grown-ups and asked them if it frightened them.

They said: 'Why should we be frightened of a hat?'

My drawing wasn't a hat. It was a boa-constrictor digesting an elephant. So I did a drawing of the inside of the boa-constrictor to help the grown-ups to understand. They always need explanations. My Drawing Number 2 was like this:

The grown-ups told me to put my drawings of open and closed boa-constrictors away and get on with my geography, history, arithmetic and grammar. And that's how I gave up a fantastic career as a painter at the age of six. I was depressed by the failure of Drawing Number 1 and of Drawing Number 2. Grown-ups never understand anything on their own and it's boring for children to keep having to explain things to them.

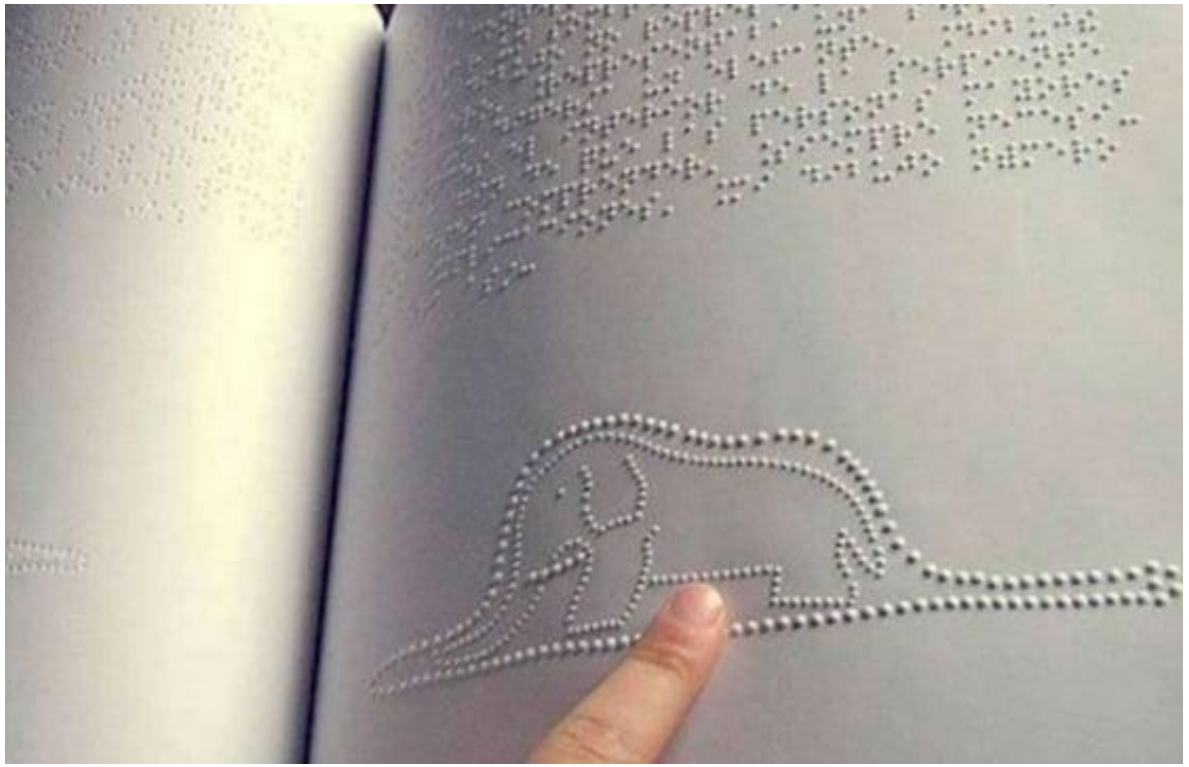
So I had to choose a different job and I learned how to fly planes instead. I've flown all over the world. And it's true that geography has been very useful. I can tell the difference between China and Arizona at a glance. This is very helpful if you get lost in the night.

In the course of this life I've had lots to do with lots of serious people. I've spent a lot of time with grown-ups.

6

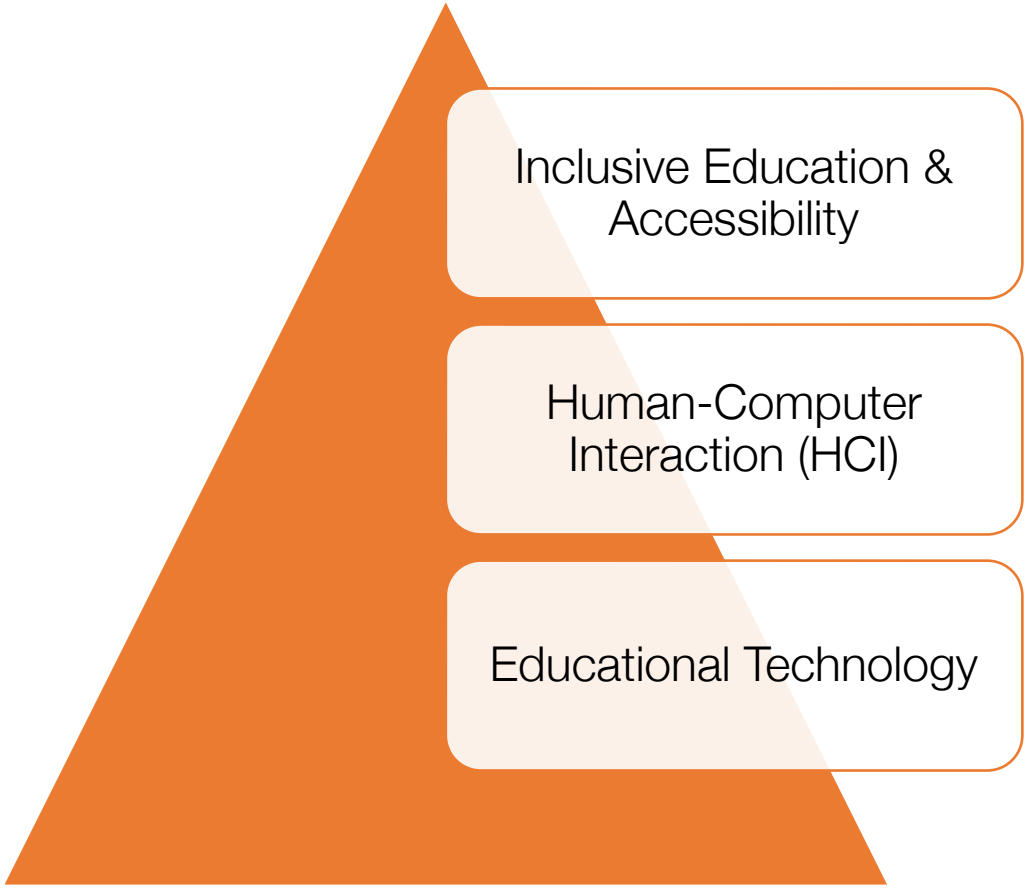
Visual elements (illustrations, colors, symbols) build early mental models and spatial relationships.

Current Materials for BVI Children



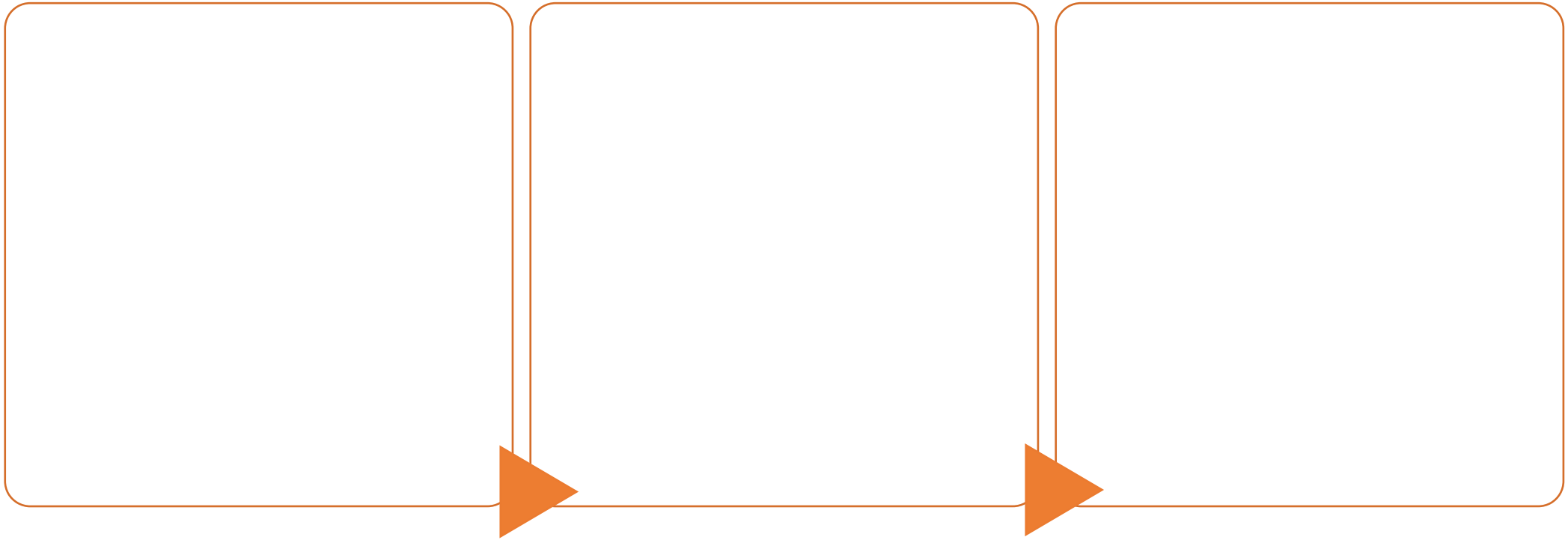
Braille provides vital textual access, but completely misses the contextual meaning embedded in visual graphics.

Existing multisensory innovations remain fragmented across multiple academic disciplines



The Core Question:
How are visual concepts represented and communicated to BVI children through alternative sensory methods?

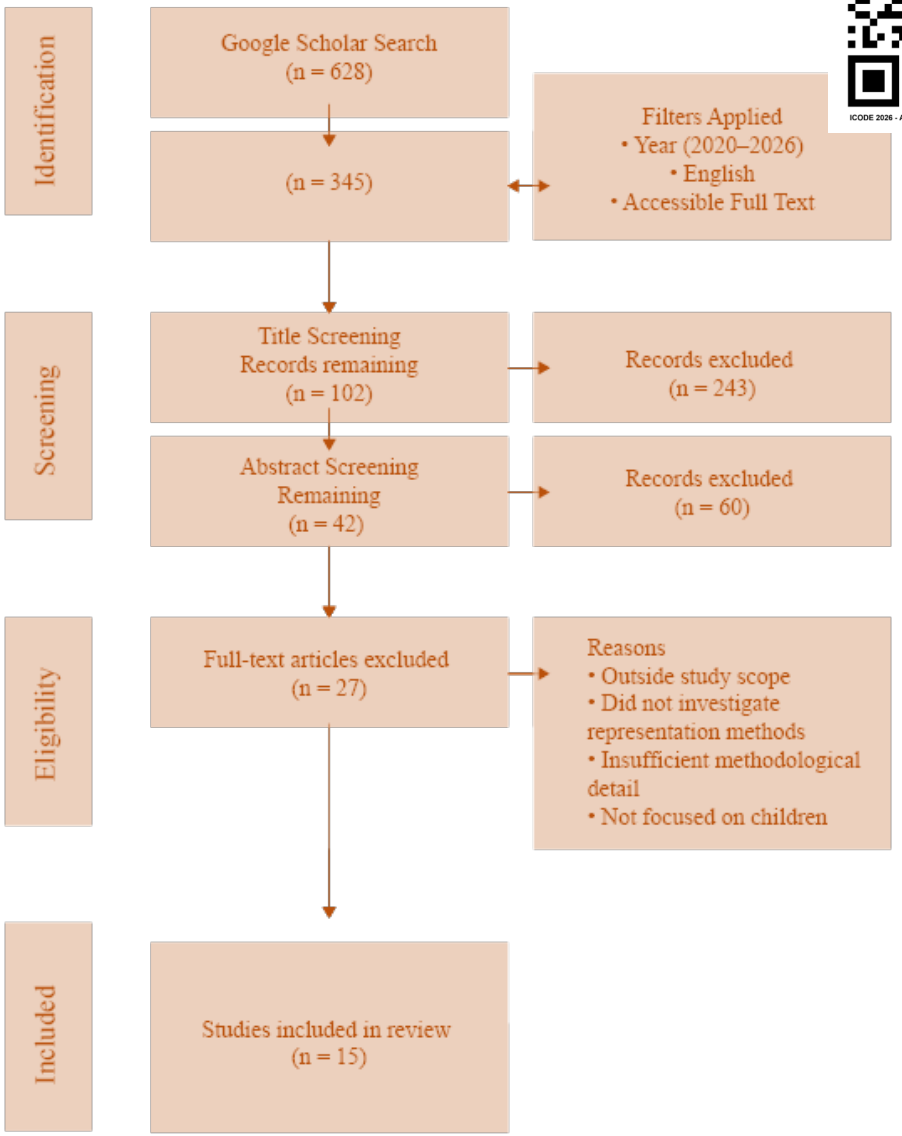
Synthesizing recent evidence to identify effective representatic strategies



Distilling the landscape of multisensory educational research down to 15 core studies

TABLE I. SYNTHESIS OF OETHODS, TECHNIQUES, MEDIA, AND DESIGN PRINCIPLES FOR REPRESENTING VISUAL CONCEPTS

Theme	Summary of Findings	Supporting Studies
Visual Concepts	Objects, environments, spatial relationships, emotions, and scientific and mathematical concepts were represented through multisensory approaches.	[1], [2], [3], [4], [8]
Representation Methods	Tactile graphics, 3D models, Braille, textures, tangible objects, sensory substitution.	[3], [5], [6], [7], [8], [10], [11
Communication Techniques	Storytelling, narration, sonification, interactive exploration, participatory design.	[1], [2], [3], [5], [9], [10]
Educational Media	Multisensory storybooks, tactile books, learning kits, digital learning platforms, VR.	[1], [5], [4], [6], [12], [13]
Sensory Modalities	Touch was dominant, supported by audio, movement, residual vision, and occasionally smell.	[1], [2], [3], [4], [5], [7], [8], [10], [16],
Technologies	3D printing, NFC, QR codes, interactive audio, VR, digital platforms.	[3], [4], [5], [6], [9], [11], [12], [13]
Design Principles	Simplification, multimodal integration, user-centred design, independent exploration, inclusion.	[1], [3], [4], [5], [7], [9], [10], [12]



Effective multisensory design relies on an integrated hierarchy o senses

Residual Vision

- For children with low vision, high-contrast colors and enlarged graphics provide supplementary cues.

Audio & Digital

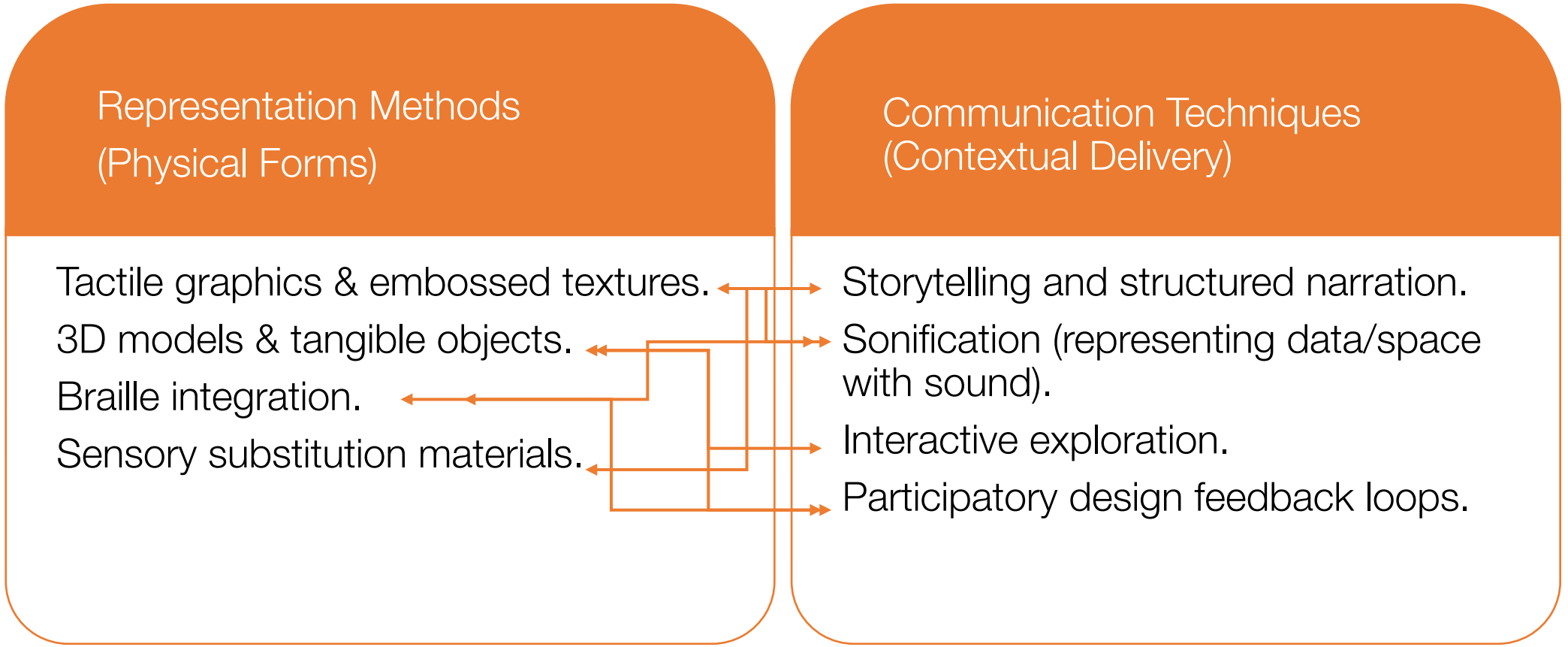
- Contextual support. Narration and sonification enhance independent exploration and reduce ambiguity.

Touch

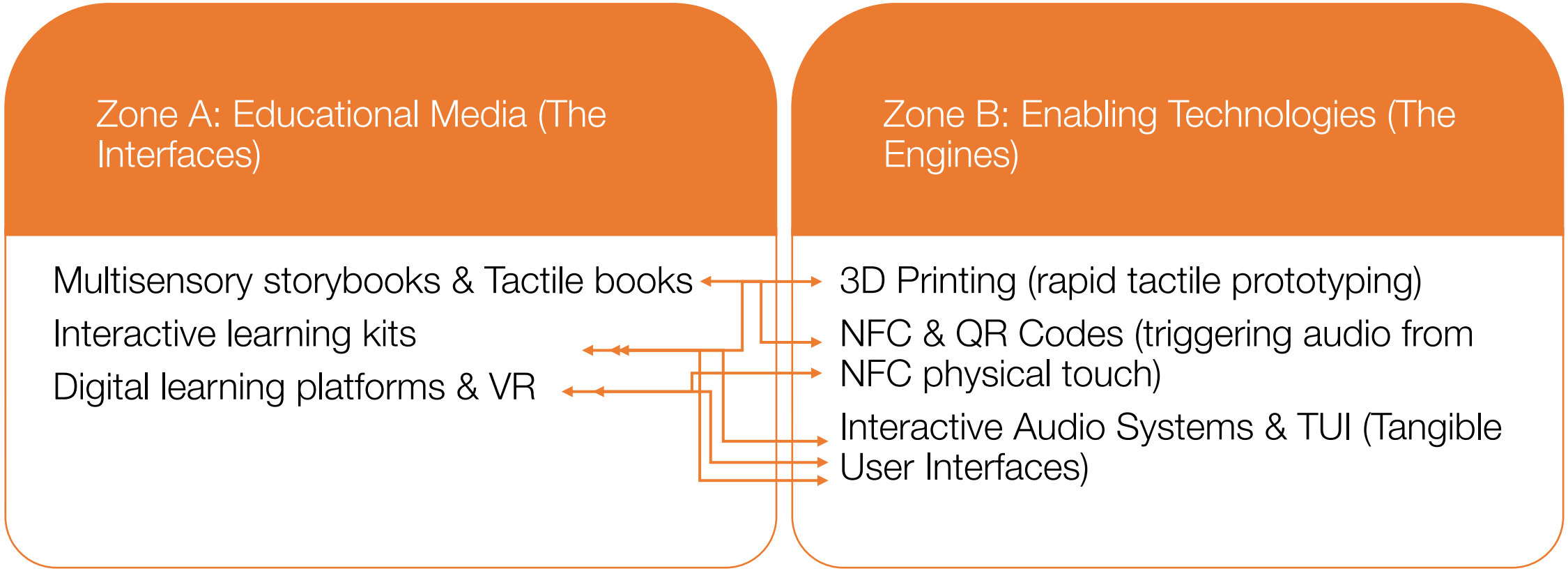
- The foundational medium. Embossed illustrations, 3D models, textures, and tangible objects communicate shape and spatial relationships.

Concept understanding requires complementary modalities working together, rather than single sensory channels.

Translating visual concepts through tactile representation and narrative context

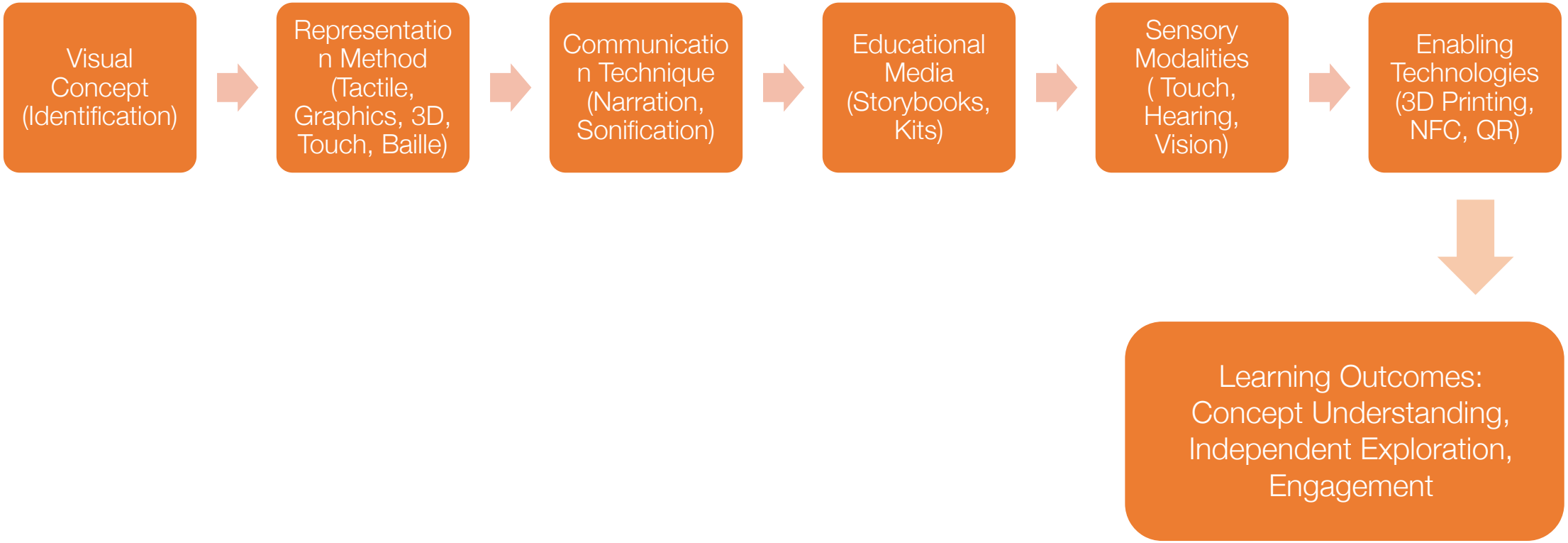


Bridging physical exploration with digital context through interactive media



Technology is not the determinant of learning outcomes; its effectiveness depends on supporting meaningful multisensory communication.

The Representation Pipeline:
A systemic approach to inclusive design.



Four foundational pillars for designing holistic learning experiences.

1	2	3	4
• **Simplification:** Distilling complex visual information into highly legible tactile forms without cognitive overload.	• **Multimodal Integration:** Layering touch, audio, and technology so senses complement rather than compete.	• **Participatory Design:** Co-designing directly with BVI children to ensure real-world usability and intuitive navigation.	• **Independent Exploration:** Designing interfaces that allow children to discover and learn without continuous sighted assistance.

Moving beyond isolated accessibility features to holistic storytelling

Accessible storytelling requires a paradigm shift. We must design **around learning concepts-integrating tactile, auditory, and digital mediums as interconnected ecosystems**—rather than appending accessibility features as an afterthought.

Select technologies for educational purpose, not novelty.

Prioritize equitable learning opportunities and independent exploration.

Thank you for your attention.